Please refer to **Appendix A** pp **A2-A6** of the recommended text book.

1. Unit 1.1

Numbers, Intervals and Inequalities

Learning Objectives

ON completion of this unit you should

(1) know the definitions of the following number systems: the set $\mathbb{N}$ of all natural numbers, the set $\mathbb{Z}$ of all integers, the set $\mathbb{Q}$ of all rational numbers, the set $\mathbb{I}$ of all irrational numbers and the set $\mathbb{R}$ of real numbers;

(2) be able to use notations;

(3) be able to use the word "and" for intersection of two sets;

(4) be able to use the word "or" for union of two sets;

(5) be able to find the union/intersection of two sets;

(6) be able to use interval notation;

(7) know the rules for inequalities and be able to use it;

(8) be able to solve inequalities.

Numbers. (please refer to page A2-A3)

Natural Numbers ($\mathbb{N}$):

Integers ($\mathbb{Z}$):

Rational Numbers ($\mathbb{Q}$):

Irrational Numbers ($\mathbb{I}$): For example $\sqrt{2}$, $\sqrt[3]{7}$, $\sin 5$, π, e, ... are irrational numbers.

Numbers (Real numbers ($\mathbb{R}$)) : In general, when we say numbers we are referring to the set of real numbers; which is the union of the set of rational numbers and the set of irrational numbers.

Exmaples

- $\frac{1}{2} =$
- $\frac{1}{3} =$
- $\frac{157}{495} =$
- $\sqrt{3} =$
- $\sqrt[3]{9} =$
- $\pi =$

The number line: A number line is used to represent real numbers. Every point on a number line corresponds to some real number.

Set: Set is a collection of well defined objects, and these objects are called elements of the set.

Set description:

- Complete listing method

- Set builder method

Union and intersection of sets: (carefully observe how we use the connectives *and* and *or*.)

Union : ($\cup$)

Intersection : ($\cap$)

The empty set: ($\emptyset$)

Intervals

Types of intervals and their geometric representation.
Let a, $b \in \mathbb{R}$ such that $a < b$.

Open interval :

Closed interval :

Half open or Half closed interval :

Infinite interval :

Combining intervals using $\cup$ and $\cap$.

Find the union and intersection of the following intervals and put your final answer in interval form

(1) $(1, 3)$, $(-2, 2)$

(2) $[-8, 3)$, $[-2, 10]$

(3) $(1, \infty)$, $(-\infty, -5]$

Inequalities

Rules of inequalities. (Please refer to page A4)

Examples: Use the rules of inequalities to solve the following and write your answers in terms of intervals

(1) $2x + 7 > 3$

(2) $1 + 5x > 5 - 3x$

(3) $1 < 3x + 4 \leq 16$

(4) $(x + 1)(x - 2) \geq 0$

(5) $x^2 + x < 2$

(6) $\frac{x+1}{x+2} \leq 0$

(7) $\frac{1}{x+1} > 1$

(8) $x^2 > 4$

(9) $x^2 < 4$

(10) $-x^2 - x - 1 < 0$